Assignment 1: Robot Kinematics and Simple Control

NOTE: To do this assignment, you will need to install PyBullet. You can find it here - https://pybullet.org/wordpress/.

Consider a three rigid-linked robot with link lengths l_1, l_2, l_3. The arm is considered to be fully outstretched and straight along the x-axis when the joint angles $\{\theta_1, \theta_2, \theta_3\}$ = 0,0,0 with the end effector (ee) pointing away from the rest of the arm. Let the robot move on a plane x,y (where z is the opposite direction of gravity)

Joint	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	l_1	0	θ_2
3	0	l_2	0	θ_3
ee	0	l_3	0	0

Table 1: D-H parameters for the manipulator

Joint	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	l_1	0	θ_1
2	0	l_2	0	θ_2
ee	0	l_3	0	θ_3

Table 2: Modified D-H parameters for the manipulator

1. Draw a figure for the arm, attach body-fixed coordinate frames, and label the coordinate frames.

2. Write the kinematic equations for the end effector position (X and Y location) and orientation (θ) in terms of sums of sines, cosines, and joint angles. Your answer should provide a separate equation for each X, Y, and θ, not a matrix.

3. We provide a Unified Robot Description Format (URDF) file to describe this robot with $l_1 = 0.5$, $l_2 = 0.5$m, $l_3 = 0.2$m. Import the robot in PyBullet and take a clear image of the arm at the following joint configurations. **Make sure all 3 links are visible.**

Configurations	θ_1	θ_2	θ_3
1	0	0	$\pi/2$
2	0	π	0
3	$\pi/2$	$\pi/2$	$\pi/4$
4	$\pi/3$	$\pi/2$	0

4. Draw the control block diagram for the robot given an input $\{x, y, \theta\}_{ref}$ for the end effector. Draw both the open-loop and the closed-loop control block diagram.

5. Describe a 2nd-order smooth profile for moving from a start location $\{x, y, \theta\}_0$ to another $\{x, y, \theta\}_{ref}$. Create an animation(.gif) of the robot moving slowly to the following locations using open-loop setpoint control. Move through locations 1-4 and return to location 1. Ensure the end effector moves straight lines between locations, reaching velocity 0 at each setpoint. Make the animation approximately 10 seconds in length. Plot the path of the end effector as it moves.

Locations	x	y	θ
1	1.2	0	0
2	0.5	0.5	0
3	0.5	-0.5	0
4	0.4	0.	0

6. Follow the same x, y trajectory of 5 using closed-loop control, but now the end effector orientation θ may be any angle. Define a nullspace controller that maximizes the angle between sequential links while still following the x, y trajectory exactly. Make an animation(.gif) as in 5.